Evidence of system

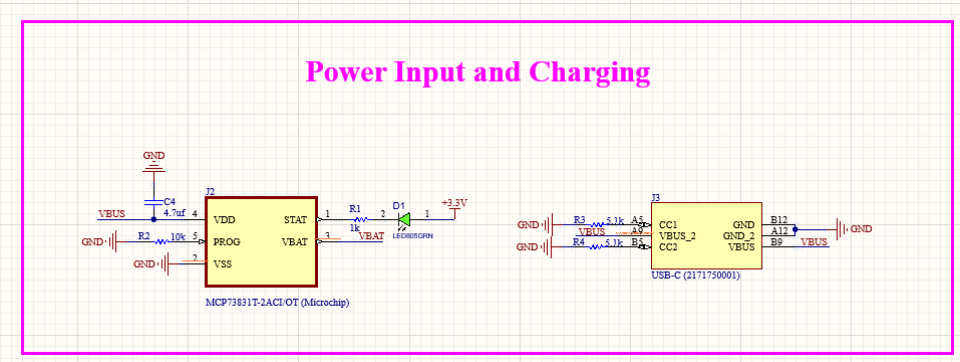
Part A. Hardware design

System

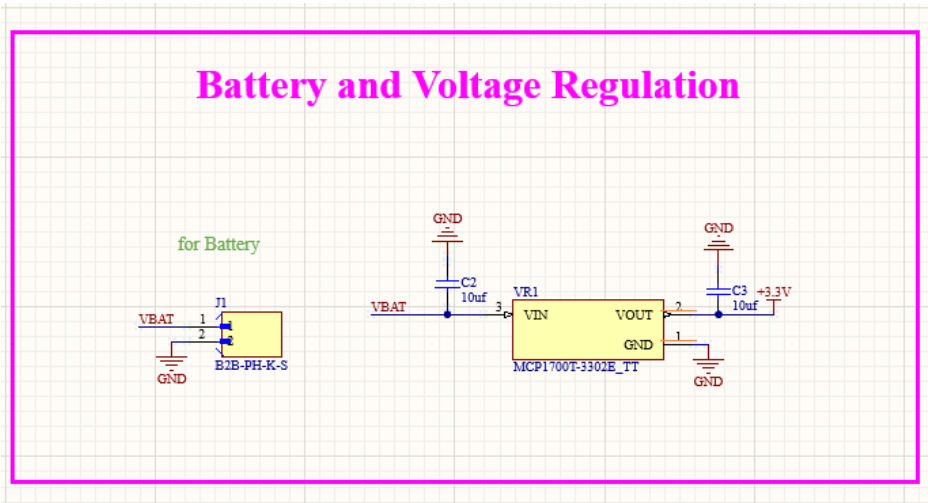


System architecture

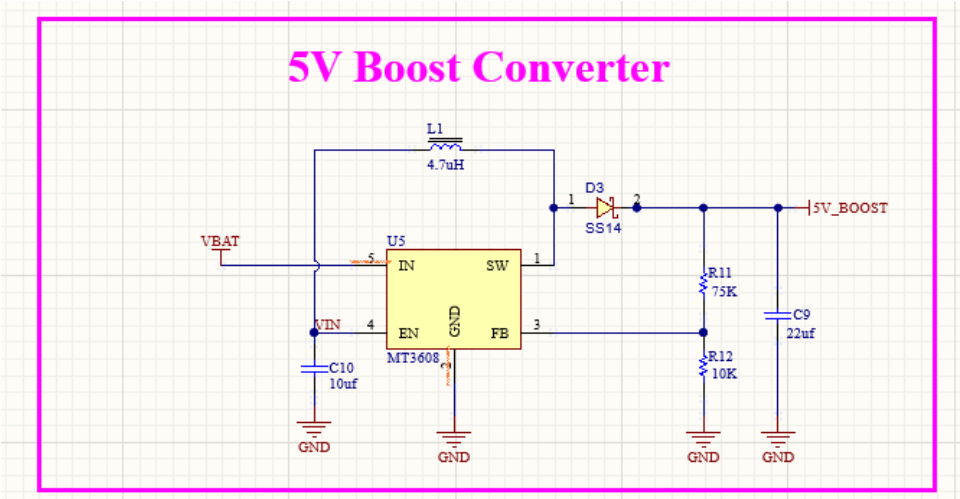
Schematic



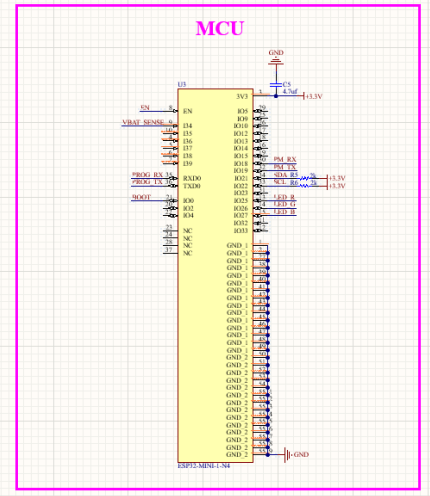
Power input and charging



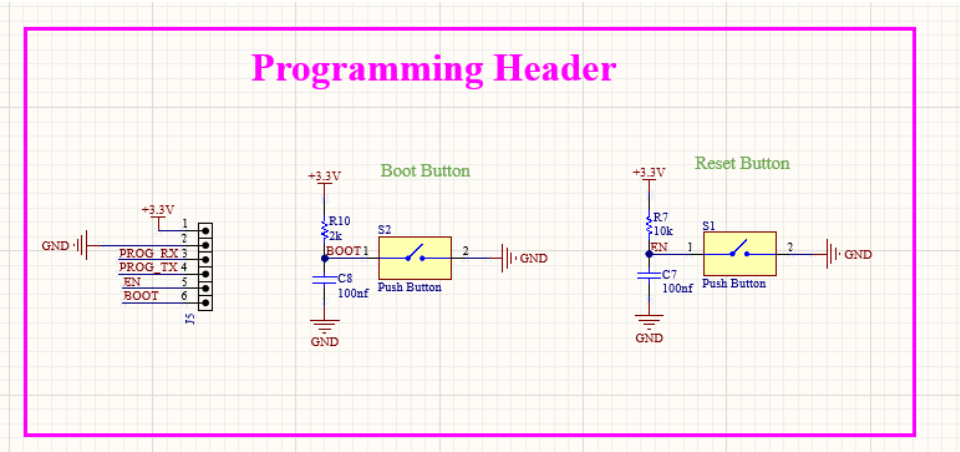
Battery and voltage regulation



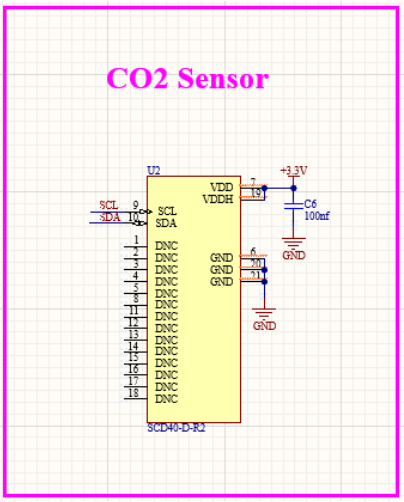
5 V boost converter



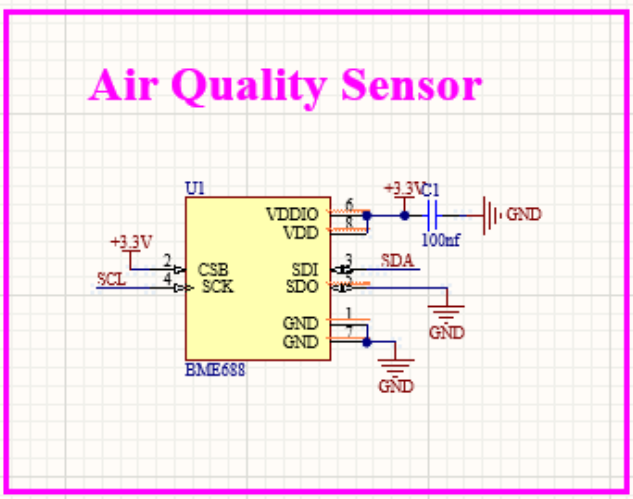
Microcontroller



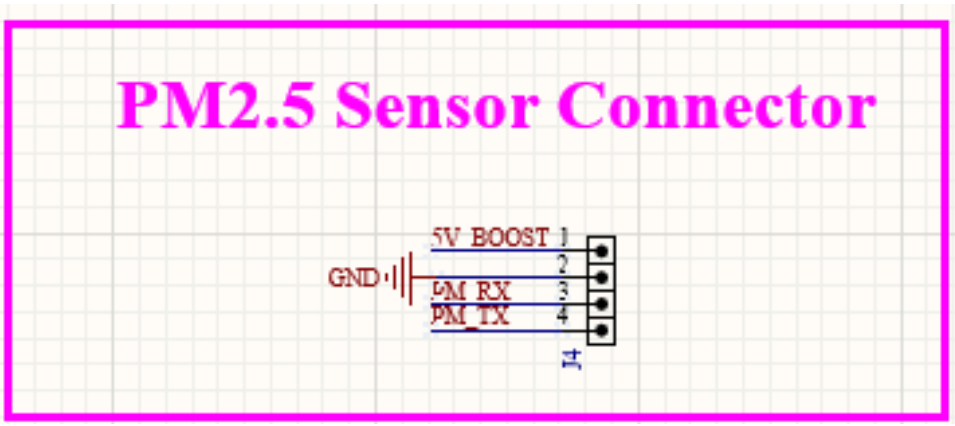
Programming header



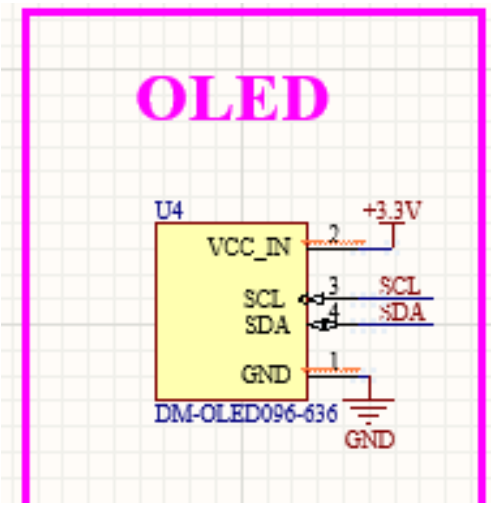
CO2 sensor



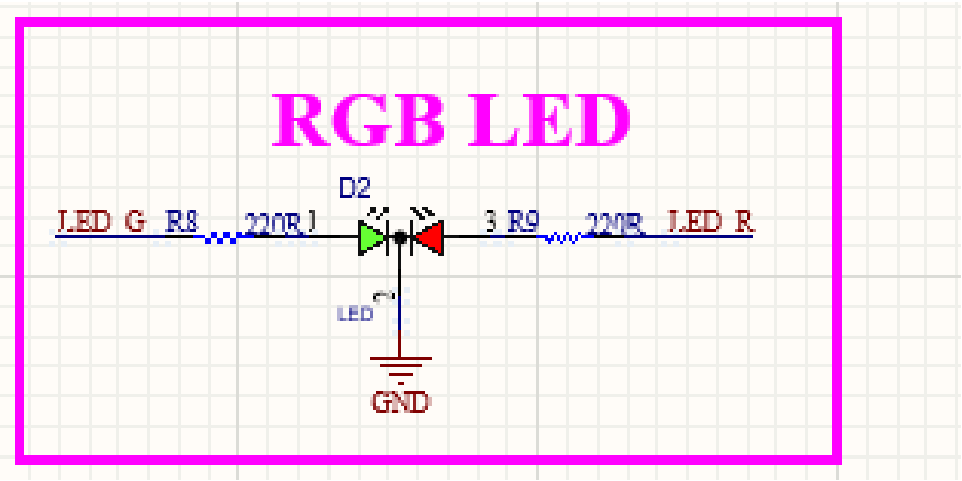
Air quality sensor



PM2.5 sensor connector

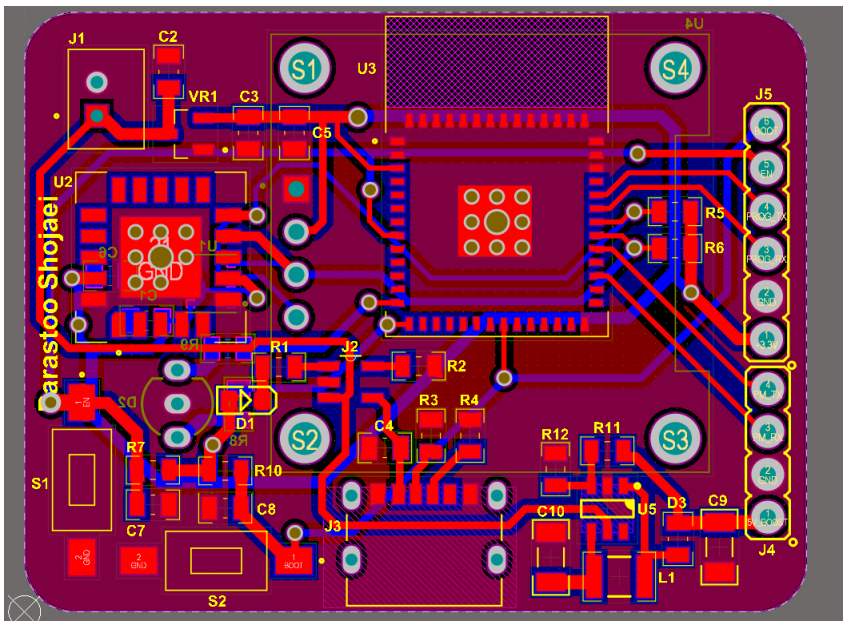


OLED display

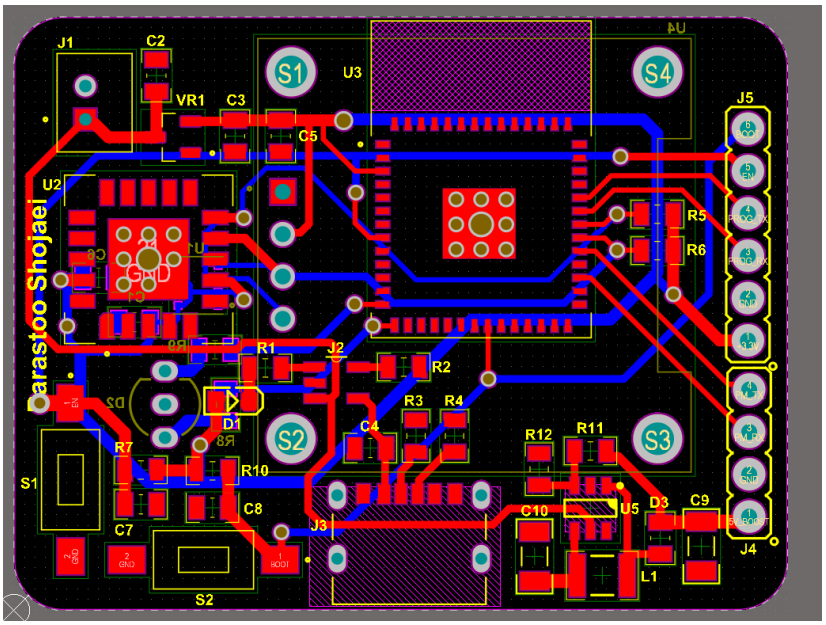


Status LED

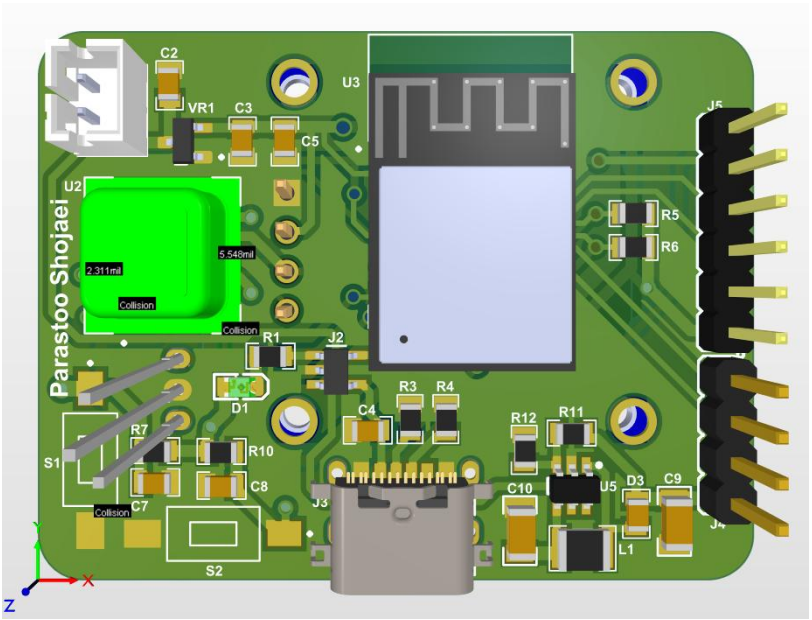
PCB layout and design check



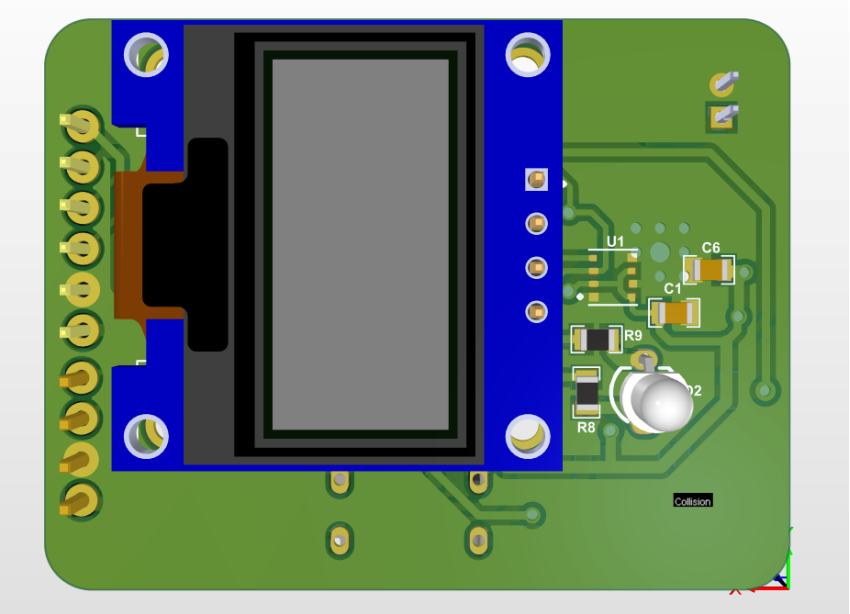
PCB layout, 2D



PCB layout, 2D



PCB, 3D top side



PCB, 3D bottom side

Altium Designer

Design Rule Verification Report

Date:	9/19/2026		
Time:	6:24:11 PM	**Warnings:**	0
Elapsed Time:	00:00:01	**Rule Violations:**	0
Filename:	E:\Parastoo\STU\Uni\Msc\Dissertation\PCB\Project\PCB.PcbDoc		

Summary

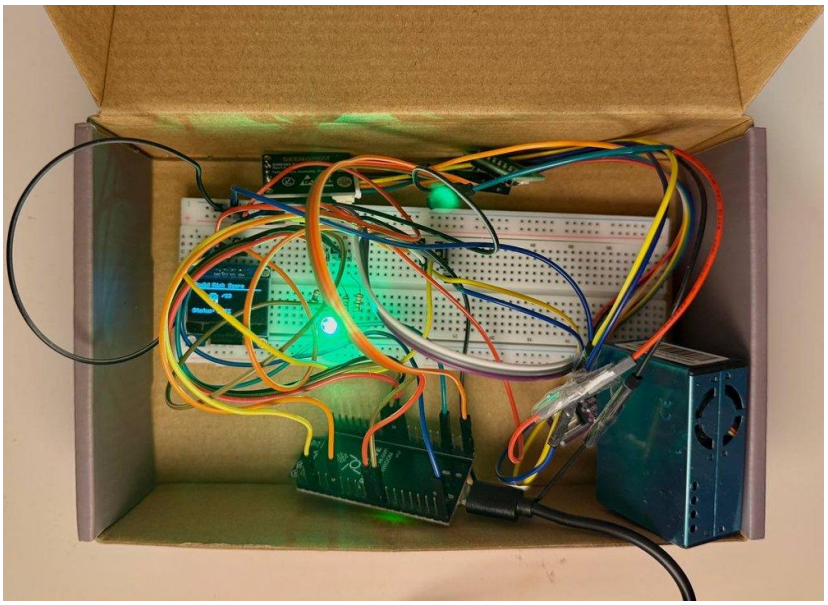
Warnings	Count
Total	**0**

Rule Violations	Count
Clearance Constraint (Gap=10mil) (All),(All)	0
Clearance Constraint (Gap=0mil) (All),(OnLayer('Keep-Out Layer'))	0
Clearance Constraint (Gap=14mil) (All),(InNet('GND'))	0
Short-Circuit Constraint (Allowed=No) (All),(All)	0

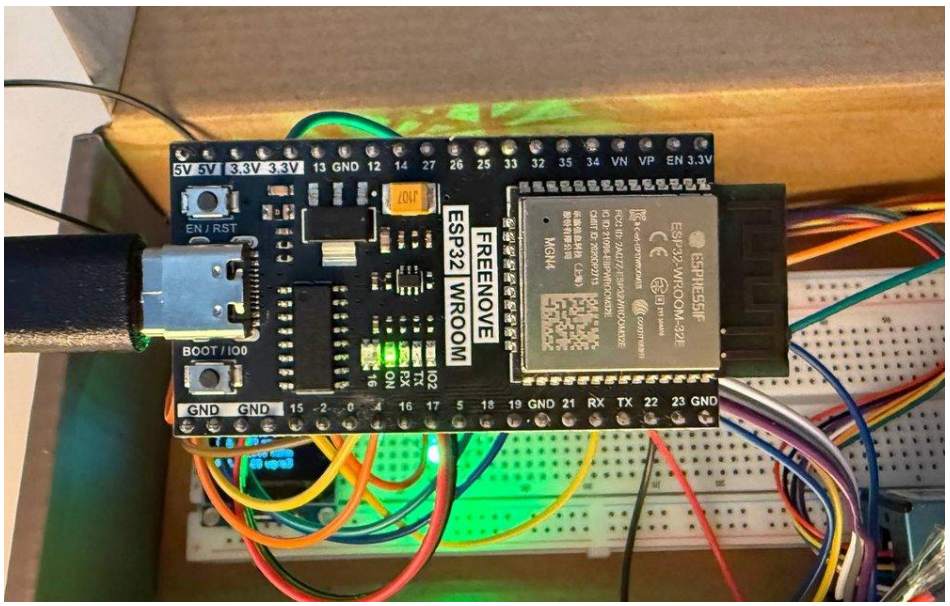
Design Rule Check report

Part B. Evidence

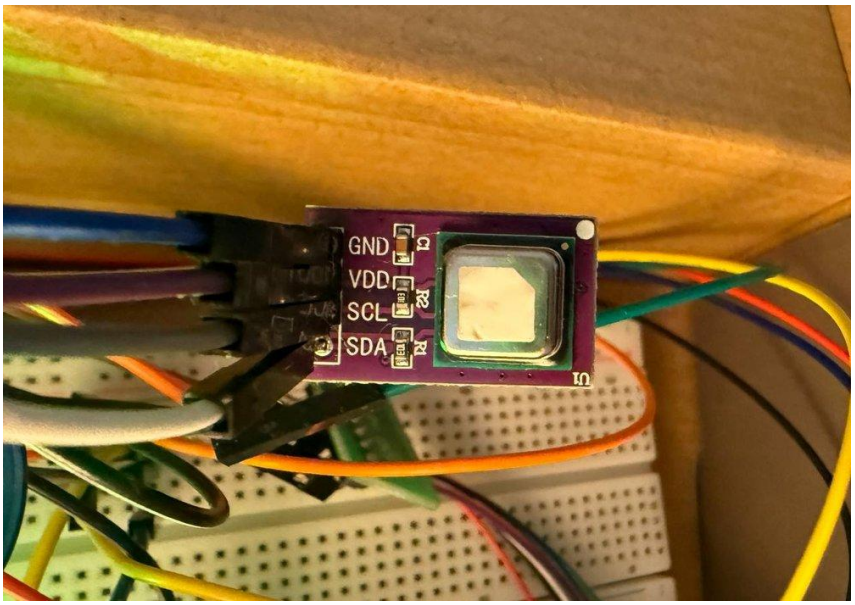
Prototype and components



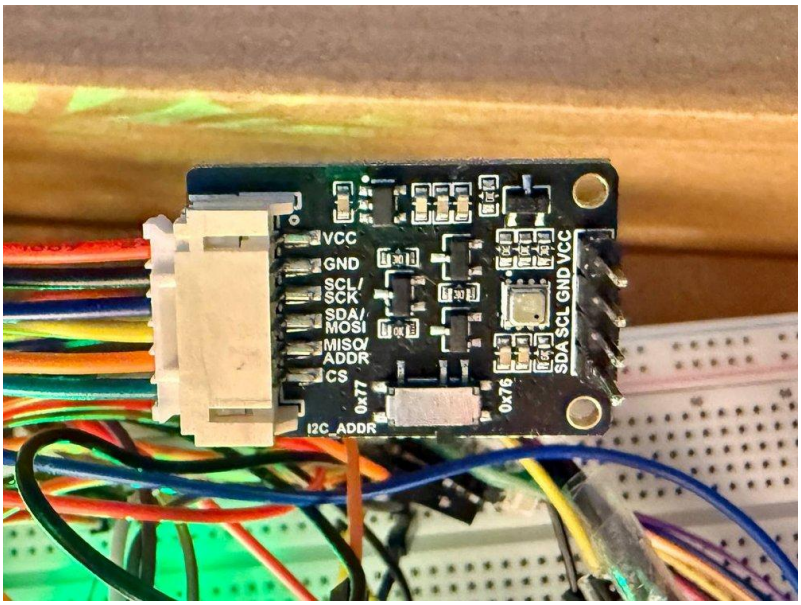
Breadboard prototype



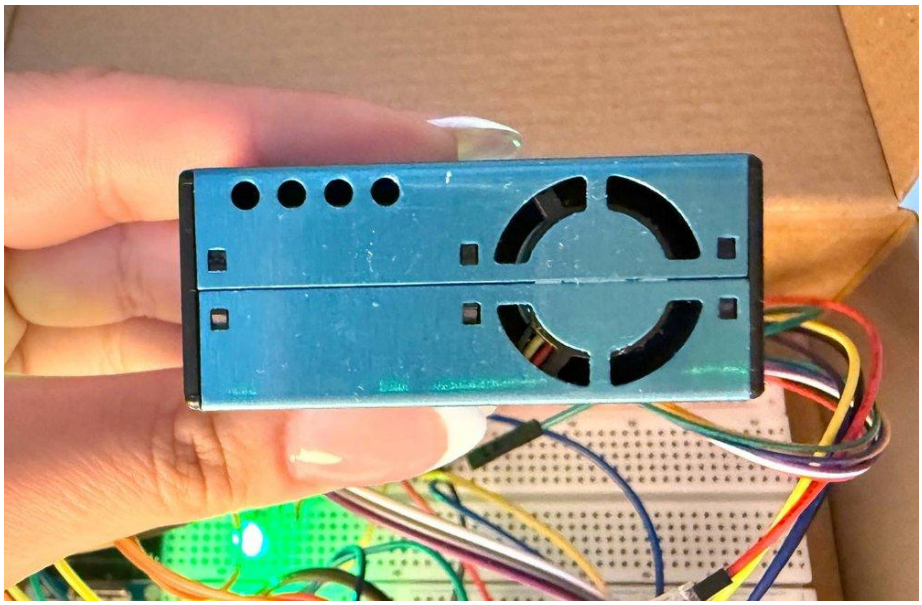
ESP32 board



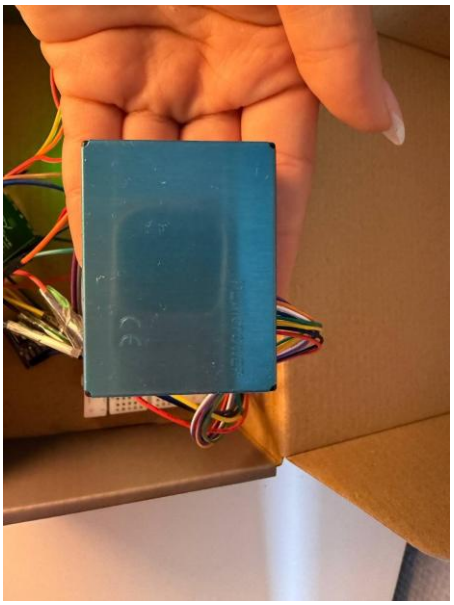
SCD40 CO2 sensor



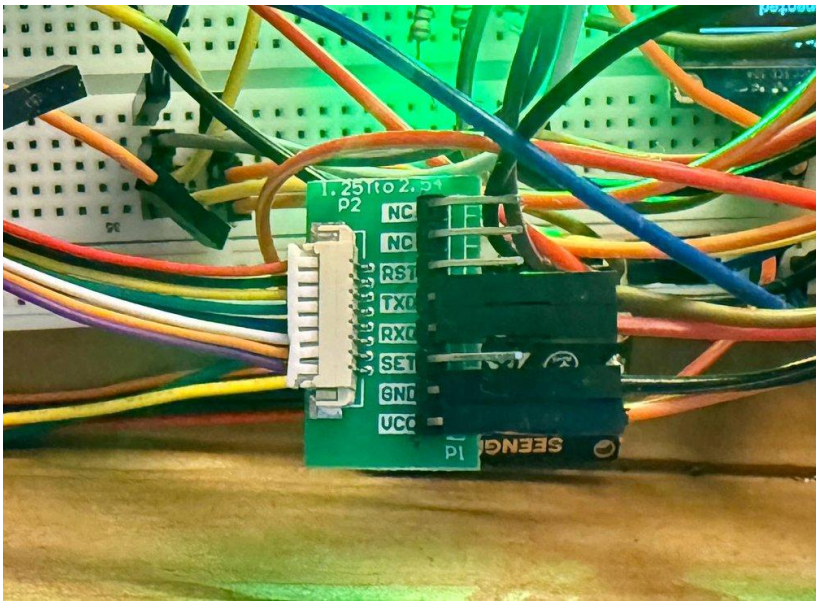
BME688 sensor



PMS5003 PM2.5 sensor



PMS5003 panel

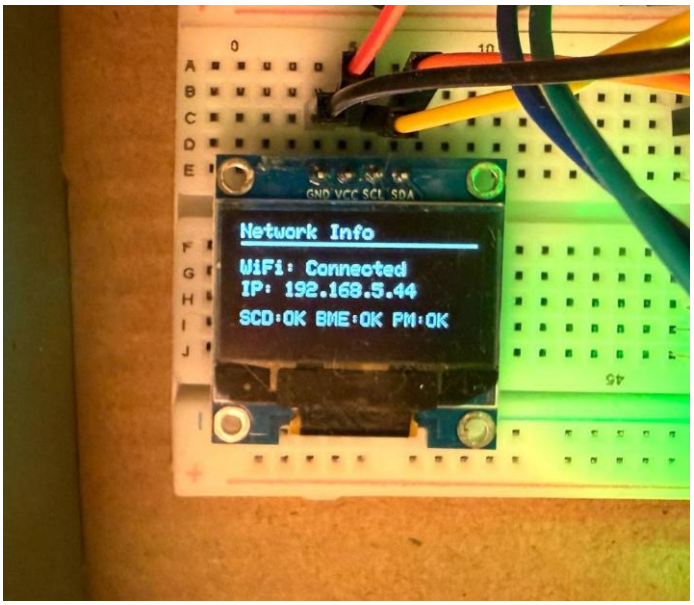


PMS5003 UART adapter

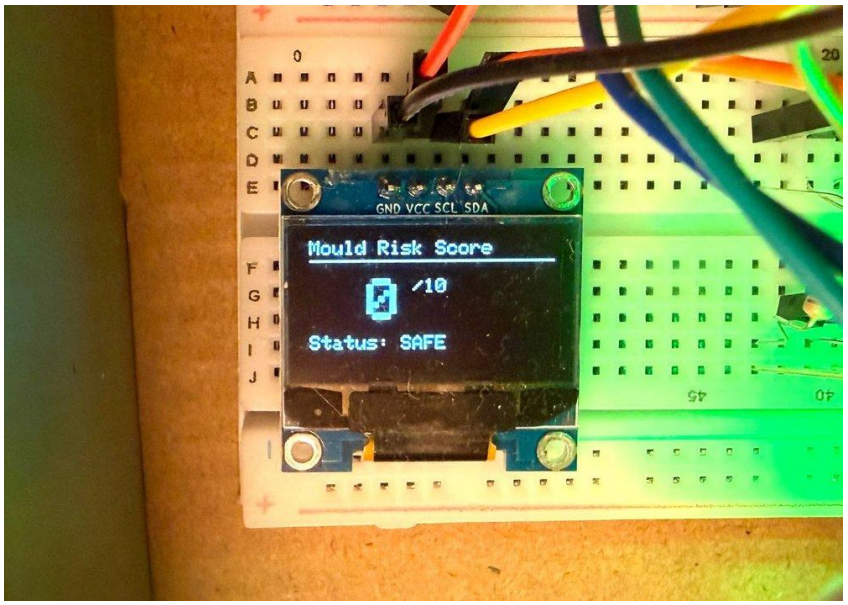
OLED display and serial output



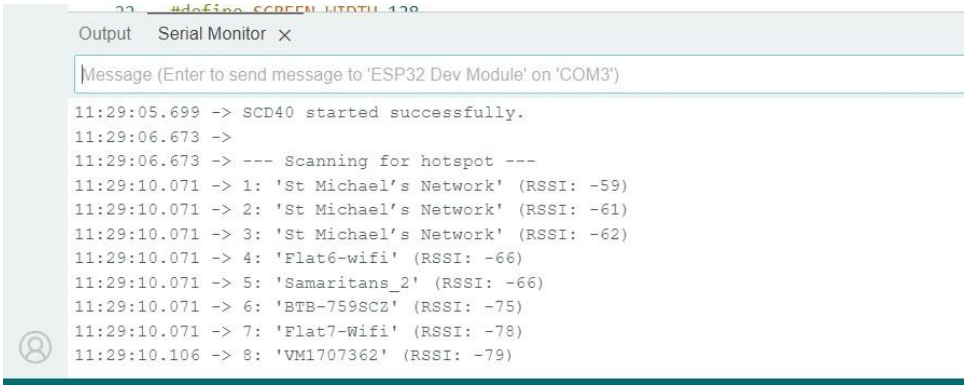
OLED readings page



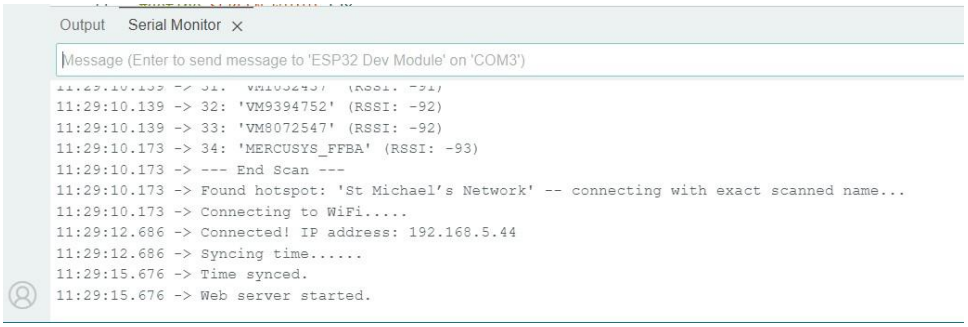
OLED network page



OLED mould-score page

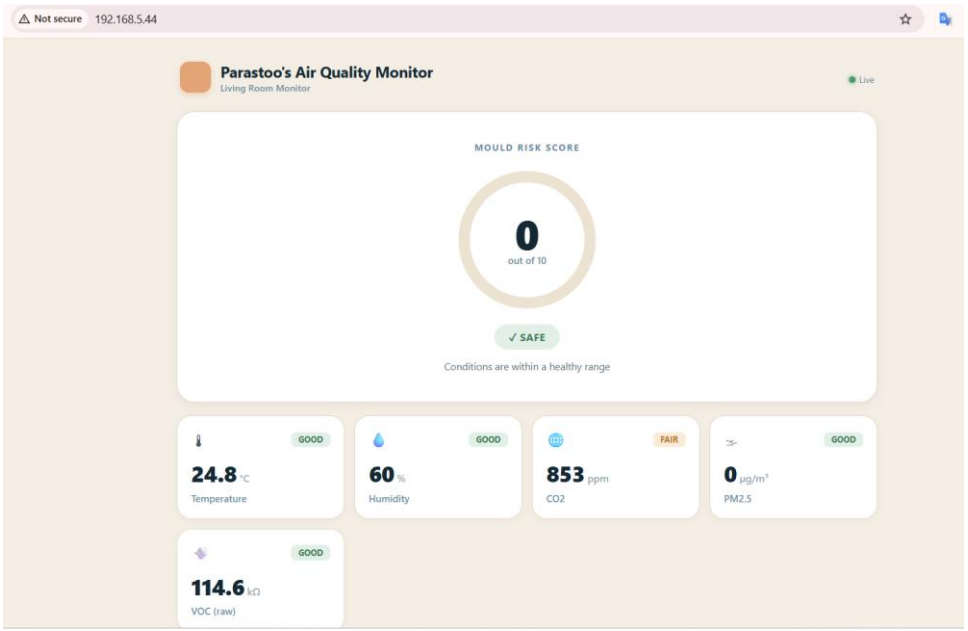


Serial console, start-up

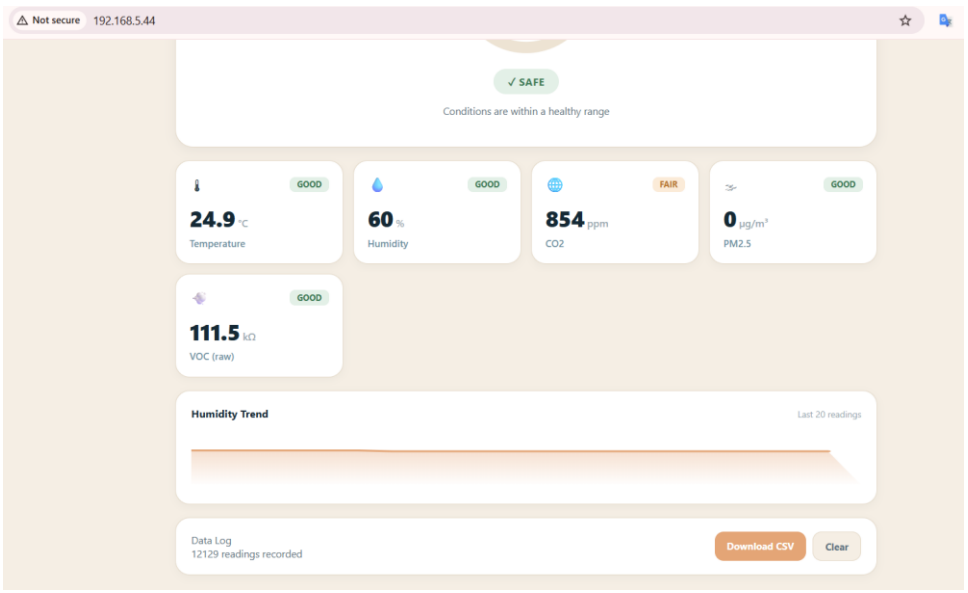


Serial console, WiFi and web server

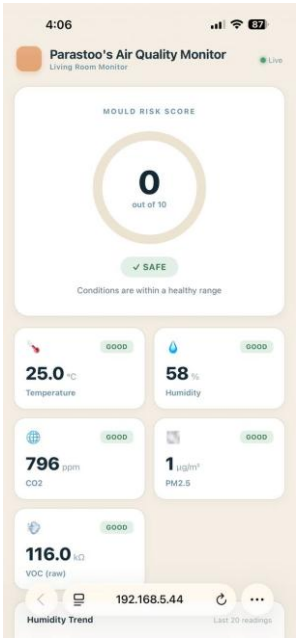
Web dashboard and data log



Dashboard, main page



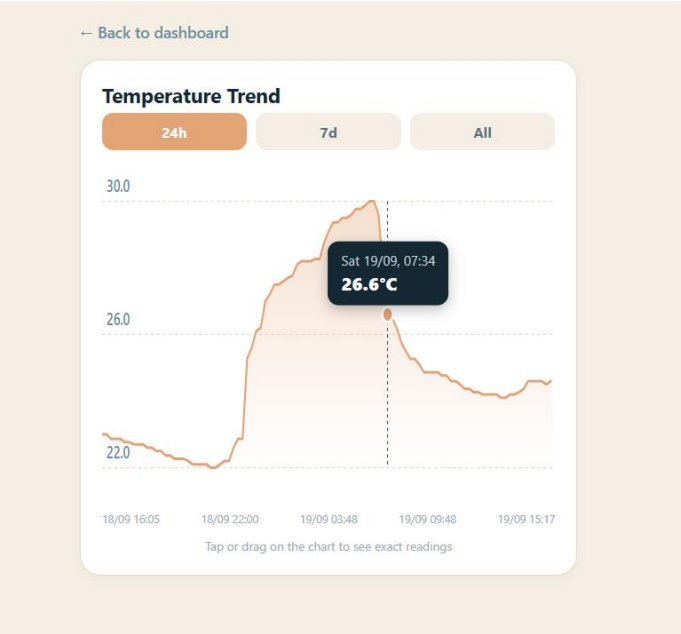
Dashboard, lower part



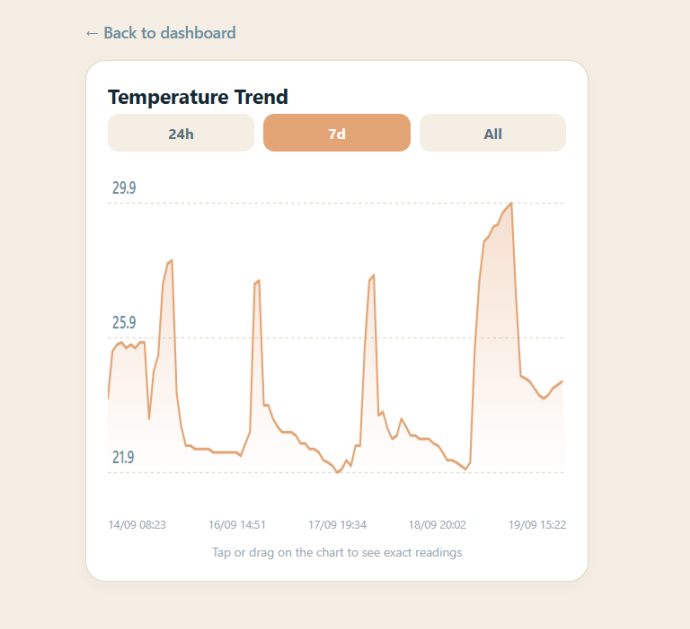
Dashboard on a phone, main page



Dashboard on a phone, trend page



Trend page, 24 hours

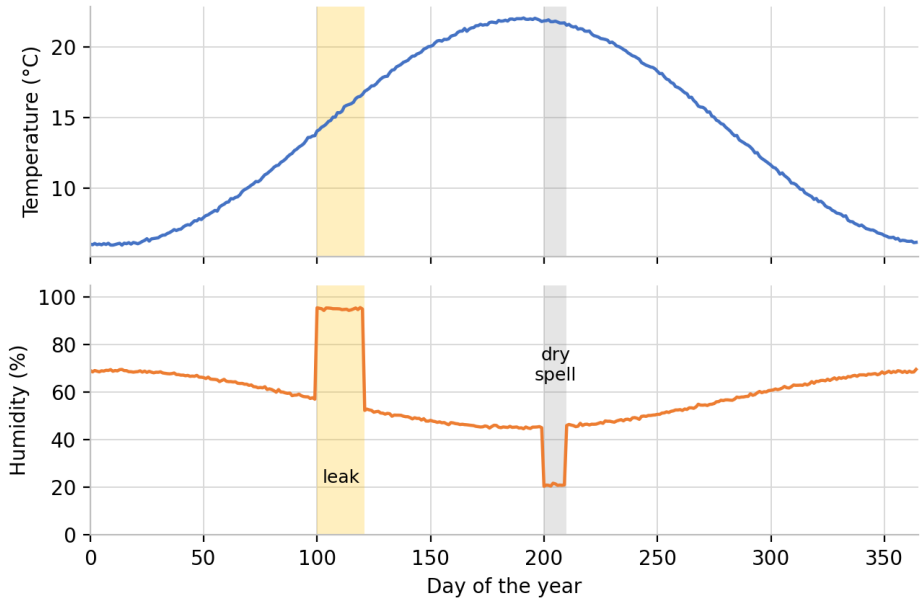


Trend page, 7 days

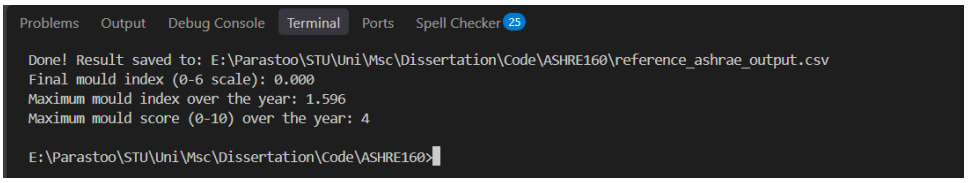
	A	B	C	D	E	F	G	H	I	J
1	timestamp	temperature	humidity_pct	co2_ppm	voc_kohm	pm1	pm2_5	pm10	mould_score	mould_index_06
2	9/14/2026 6:09	25.6	65.8	857	68.2	0	1	1	0	0
3	9/14/2026 6:10	24.8	65.7	843	67.3	0	1	2	0	0
4	9/14/2026 6:10	24.7	65.6	841	67.3	0	1	2	0	0
5	9/14/2026 6:11	24.7	65.6	842	67.9	0	1	2	0	0
6	9/14/2026 6:11	24.7	65.6	842	68	0	1	2	0	0
7	9/14/2026 6:12	24.7	65.6	844	68.2	0	1	2	0	0
8	9/14/2026 6:12	24.7	65.5	844	68.7	0	1	2	0	0
9	9/14/2026 6:13	24.7	65.5	843	68.9	0	1	2	0	0
10	9/14/2026 6:13	24.7	65.4	841	69.2	0	1	2	0	0
11	9/14/2026 6:14	24.7	65.4	841	69.5	0	1	2	0	0
12	9/14/2026 6:14	24.7	65.4	842	69.9	0	1	2	0	0
13	9/14/2026 6:15	24.7	65.5	842	69.6	0	1	2	0	0
14	9/14/2026 6:15	24.7	65.4	841	69.6	0	1	1	0	0
15	9/14/2026 6:16	24.7	65.9	842	68.9	0	1	1	0	0
16	9/14/2026 6:16	24.8	67.2	841	68.5	0	1	1	0	0

CSV log in Excel

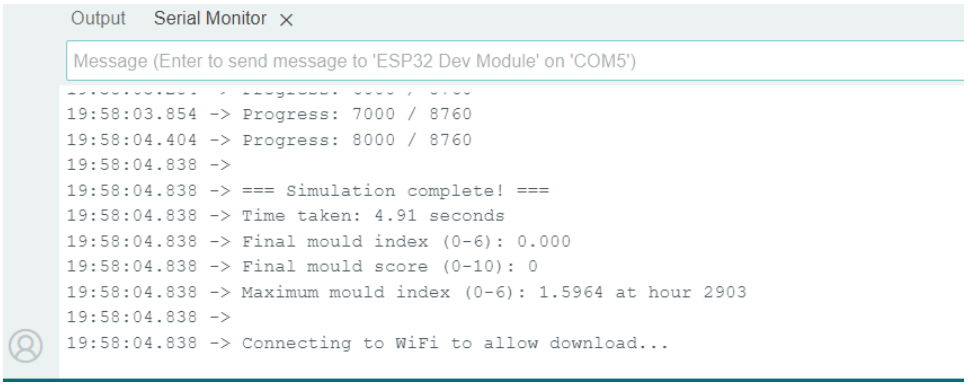
Algorithm validation



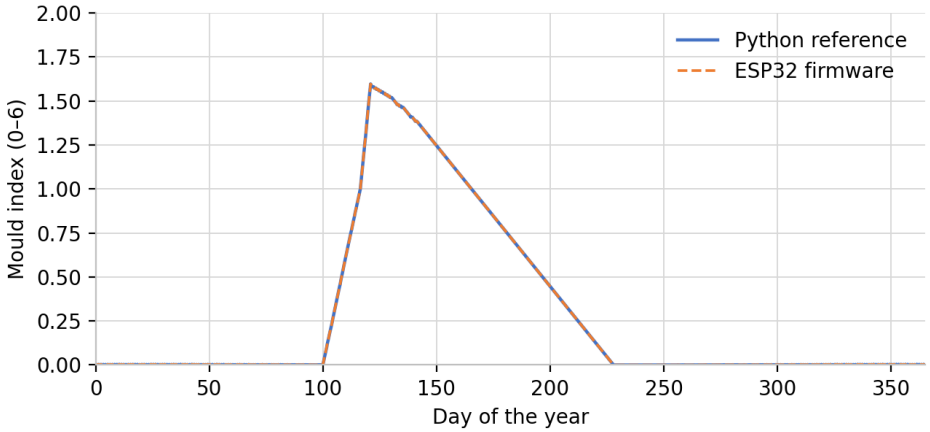
Synthetic dataset



Python reference run

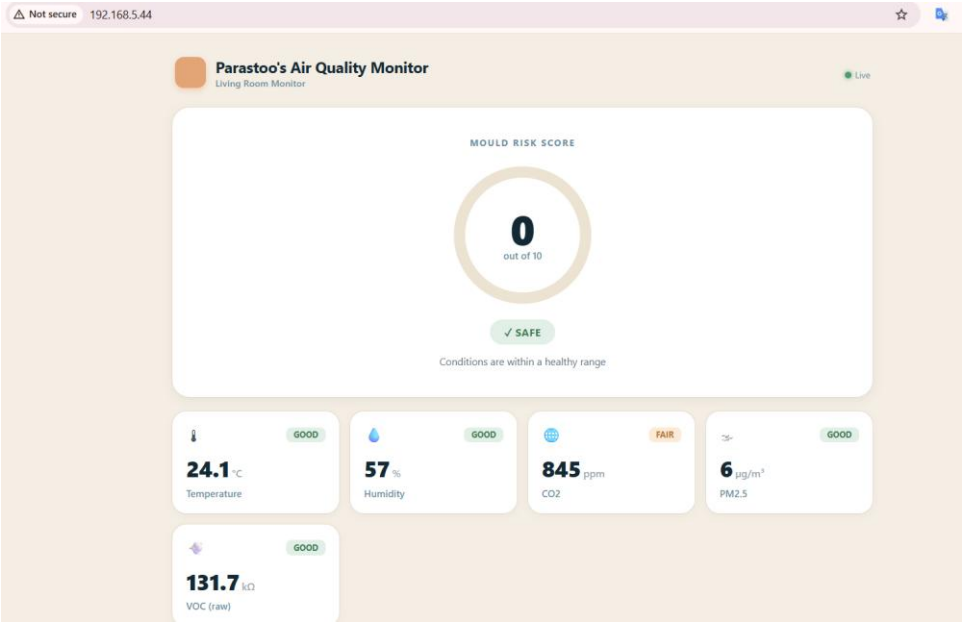


ESP32 test run

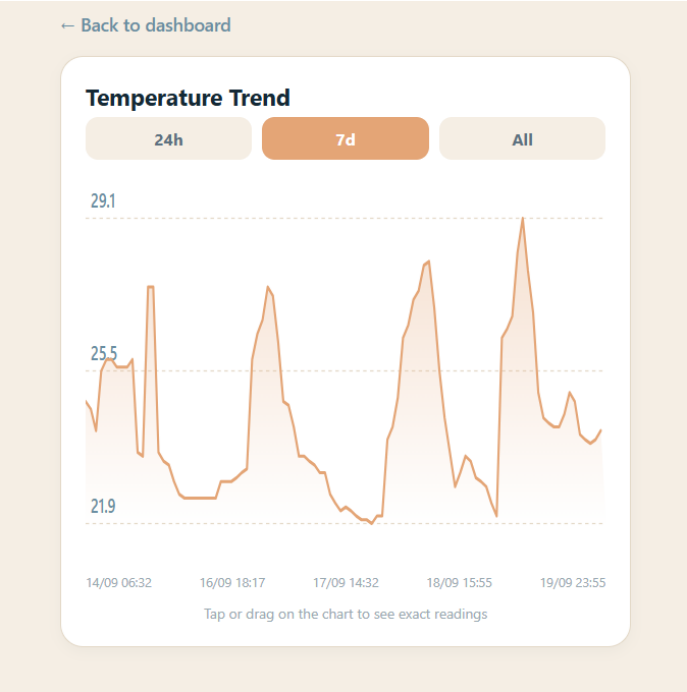


Mould index over the year

Real-world test



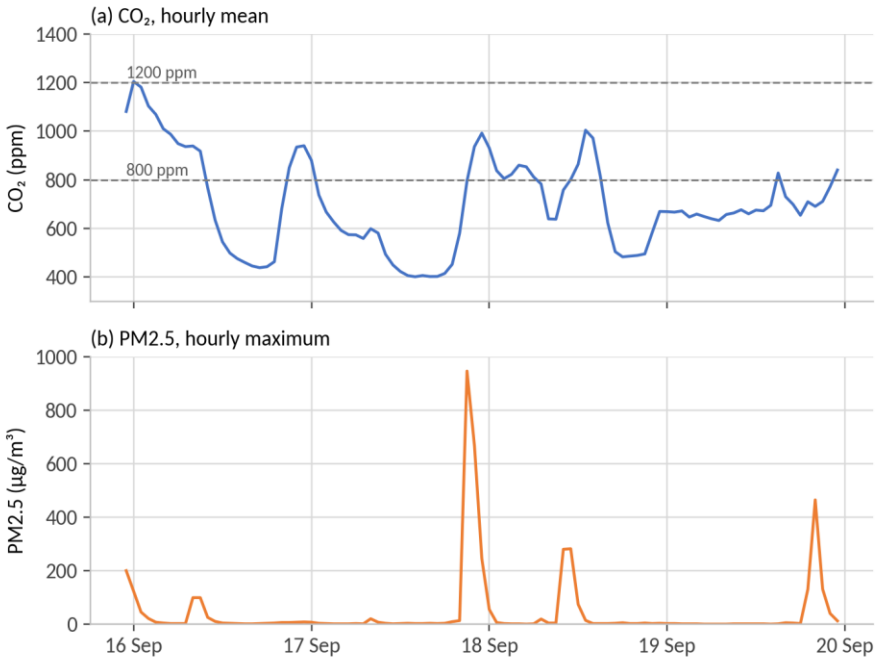
Dashboard, 19 September 23:58



Dashboard 7-day temperature

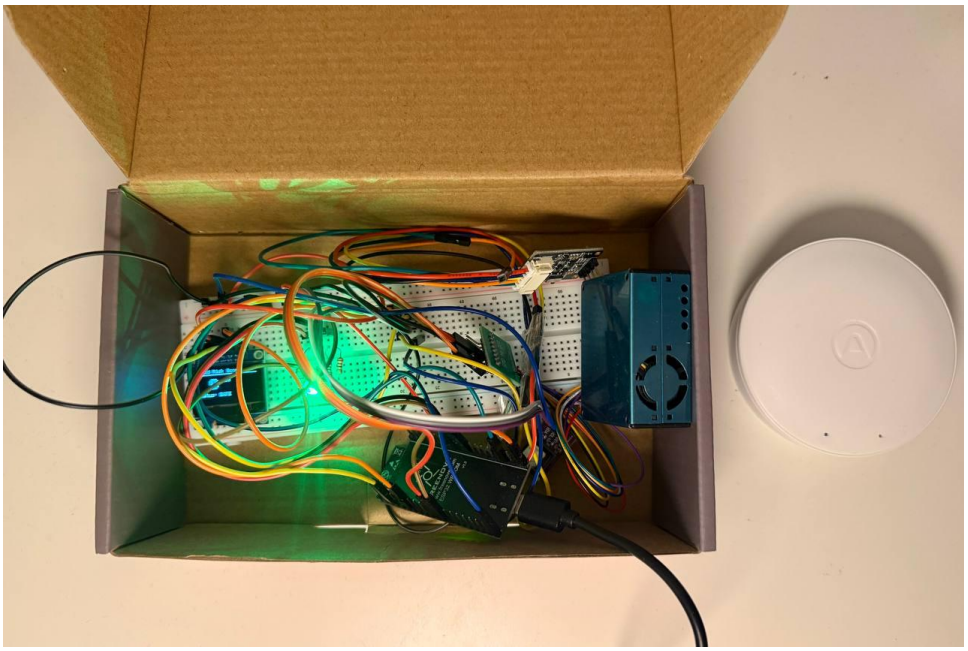


Dashboard 7-day humidity



CO2 and PM2.5, real log

Comparison with the Airthings Wave Mini



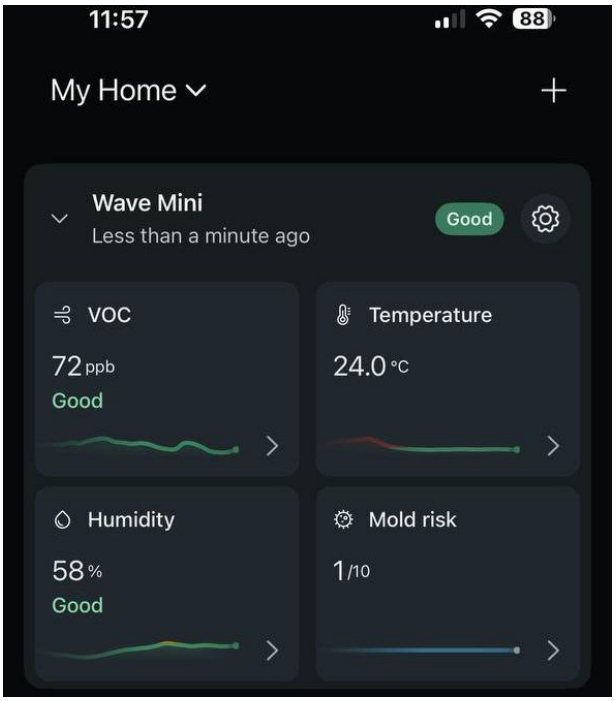
Prototype and Wave Mini



Wave Mini 7-day temperature



Wave Mini 7-day humidity



Wave Mini app, 19 September 23:57



Wave Mini colour light